



Mariners 2027 Data Science Intern Problem Set

Below is the set of problems we would like you to complete. You will be evaluated not just on your predictions, but also on your ability to communicate clearly and the quality of your process.

1. *In the Google Drive folder linked below you will find three files: `data-train.csv`, `data-test.csv`, and `data-dictionary.csv`.*

We have provided a season of pitch tracking data Minor League baseball, along with play metadata, pre-filtered to called pitches. The `train.csv` file includes two additional columns: on pitches that are challenged and reversed, `is\_challenge` will be 1, `challenge\_source` will give the unit responsible for the challenge (`hitting\_team` or `pitching\_team`), and `is\_successful` will be 1 on reverses, 0 on call confirmed. On pitches that aren't challenged, `is\_challenge` will be 0 and `challenge\_source`/`is\_successful` will be null. Challenge rules can be found [here](#).

Your objective is to predict the likelihood of a challenge in the `data-test.csv` file and fill out the `p\_challenge`, `challenge\_source`, and `p\_success\_g\_challenge` columns in that .csv with your estimates. To assist you, we have included the `data-dictionary.csv` file to explain what each column in the attached datasets represents.

*You may use whatever method or language you like, and you are encouraged to use AI coding tools. However, **you must submit the code you used to generate and evaluate your predictions**. You will be evaluated on both the **log loss** score of your predictions and on **your process in generating the predictions**. Please also include a brief explanation of your process for an R&D audience and what steps you would take to improve on this model if you had more resources. While we encourage you to use AI coding tools in your model development, **you must write the report in your own words**; we want to see how well you understand your process, and we will fail you if you use AI to provide that explanation.*

2. *Yesterday our affiliate played a game with game_id `01103572` as the home team. Generate a one-page post-game report for a coaching audience evaluating our catcher performance with respect to ABS challenges.*
3. *In approximately 300 words, and using only information available at the time of the decision, what is a recent mistake the Mariners organization has made, and why do you consider it a mistake?*

Data for problems 1 and 2 can be found [here](#).

Due to the large size of the train and test datasets, consider either zipping the file before emailing or storing it in a public cloud storage site like Google Drive or Dropbox.

If you have any questions, or need clarification of any kind, please message Forrest Diamond at fdiamond@mariners.com or John Edwards at jedwards@mariners.com.